Cheat Codes :

PHYSICAL CHEMISTRY

1.) Gibbs Free Energy & Cell Potential [2025, 2024, 2023, 2021, 2019]

$$\Delta G^\circ = -nFE^\circ_{cell}$$

2.) First Order Integrated Rate Law [2025, 2024, 2023, 2022]

$$k = \frac{2.303}{t} \log \frac{[A_0]}{[A]}$$

3.) Equilibrium Gibbs Energy [2025, 2023, 2019]

$$\Delta G^\circ = -2.303\ RT \log K_c$$

4.) Bohr's Orbit Energy [2025, 2024, 2023, 2020]

$$E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2}\ J/atom$$

5.) Mole formulae : [2024, 2023, 2022, 2020]

$$n = \frac{Wt.}{M}$$

6.) Boiling Point elevation [2025, 2024, 2021]

$$\Delta T_b = i k_b m$$

7.) Faraday's Law (Mass deposited) [2025, 2024, 2020]

$$Wt. = \frac{E \cdot I \cdot t}{96500}$$

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8.) First Order Half Life [2024, 2023, 2022]

$$t_{1/2} = \frac{0.693}{k}$$

9.) Bohr's orbit Radius [25, 2023, 22]

$$r_n = 52.9 \frac{n^2}{Z}$$

10.) Osmotic pressure [25, 24, 21]

$$\pi = iCRT$$

11.) Molar Conductivity [25, 24, 21]

S/cm

$$\Lambda_m = \frac{k \times 1000}{M}$$

12.) Rydberg formulae [25, 24, 2019]

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

13.) Solubility product [21, 20, 19]

$$k_{sp} = [xS]^x [yS]^y$$

14.) Degree of dissociation [25, 24, 21]

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$$

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15.) Energy of a Photon [2025, 24, 21]

$$E = \frac{hc}{\lambda}$$

16.) Nernst Equation (at 298 K) [25, 22, 2019]

$$E_{cell} = E^\circ_{cell} - \frac{0.059}{n} \log Q$$

17.) Arrhenius Equation (Comparison) [24, 21]

$$\log \frac{K_2}{K_1} = \frac{E_a}{2.303\,R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

18.) The K_p & K_c link: [2024]

$$K_p = K_c (RT)^{\Delta n_g}$$

19.) Raoult's law: (Vapour pressure) [23, 21]

$$P_{total} = P_A^\circ X_A + P_B^\circ X_B$$

20.) Isothermal Reversible work [24, 19]

$$W = -2.303\, nRT \log \frac{V_2}{V_1}$$

21.) pH of Salt Hydrolysis (WA + WBase) [2021]

$$pH = 7 + \frac{1}{2}(pK_a - pK_b)$$

22.) Mayers formulae : [2021]

$$C_p - C_v = R$$

23.) Henry's law [2024]

$$P = k_H \cdot X$$

24.) Entropy / enthalpy relation [23, 19]

$$\Delta G = \Delta H - T\Delta S$$

25.) Cell Constant relation [23]

$$G^* = k \cdot R$$

26.) Kohlrausch law : [2025, 2021]

$$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$$

27.) Empirical formula Calculation [24, 21]

$$\text{Atomic ratio} = \frac{\text{\% Composition}}{\text{At. Mass}}$$

28.) Limiting Reagent Calculation on eq^n basis. [24, 23]

29.) Magnetic Quantum Numbers : [2023]

$$n_m = (2l + 1)$$

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30.) Henderson- Hasselbach (Buffer) [2025, 21]

$$pH = pKa + \log \frac{[Salt]}{[Acid]}$$

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Cheat Codes

INORGANIC CHEMISTRY

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1.) VSERR : shapes & Geometry (25, 24, 22, 21, 2019)

(especially xenon Compounds – XeF_2, XeF_4)
also, SF_6, BrF_5

2.) Magnetic Moment Calculation : (25, 24, 21, 20)

$\mu = \sqrt{n(n+2)}$ BM

- Fe^{2+}
- Cr^{3+}

3.) Spectrochemical series (Learn) (25, 23, 22, 19)
($CN^- > en > NH_3 > H_2O > F^-$)

- High spin
- Low spin Complex

4.) Molecular Orbital Theory (MOT) (25, 24, 21)

Calc. Bond order $\frac{1}{2}(N_b - N_a)$

↳ They love asking → O_2 → is paramag or Not
↘ B_2 ↗

5.) Lanthanoid Contraction (25, 24, 21, 19)
↳ why Zr & Hf almost same size ?
↳ filling of 4f orbitals – affecting At. radii & Oxidn states

6.) Ionization enthalpy Trends : (25, 24, 21)
Exception – in 2nd period
why Be > B
N > O

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7.) Xenon Compound Chemistry (25, 22, 21, 19)
↳ Lone pair Count
↳ Hybridisation of · $XeOF_4$
· XeF_2

8.) Coordination isomerism (25, 24, 21)
Diff^r btw – linkage
– ionization
– solvate isomers

9.) Grp 16 Hydride Trends (24, 22, 21)
↳ why H_2O unusual high BP compared to H_2S & H_2Te (H- bonding)

10.) Qualitative Analysis (25, 24)
↳ which reagents ppt. which Cations.

11.) Crystal Field Theory (CFT) (23, 19)
↳ How d-orbitals split into t2g & eg lvls

12.) IUPAC naming (Z > 100) (easy!) (24, 22, 20)
↳ learn the roots 0 = nil
1 = un
2 = bi

13.) Oxd^n states of Mn/Cr (22, 20, 19)
· $KMnO_4$
· $K_2Cr_2O_7$
] in acidic/Alkaline Media.

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14.) Dipole Moment & polarity (24, 21, 20)

└ Identifying symmetric molecules ω zero dipole moment

- BF_3
- CO_2
- XeF_4

15.) Disproportionation rxn: (2023, 2019)

└ one element oxd & reduced at same time.

Pro Tips: – configuration of e^- from Sc to Zn
– Chelate ring forming ligands (en, EDTA)
– size → isoelectronic species (O^{2-}, F^-, Na^+)
– Salt Analysis – Colours

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Cheat Codes

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ORGANIC CHEMISTRY

1.) IUPAC Nomenclature: (2025, 2024, 2023, 2022, 2021, 2019)
↳ Mainly polyfunctional compounds
↳ Branched alkanes / alkenes

2.) Biomolecules (Glu & Amino Acids) (25, 24, 23, 21, 20, 19)

↳ Glu rxn. with HI, NH_2OH or Br_2 water
↳ Amino A. – essential & non-essential

3.) Nucleophilic Addn to Carbonyls (25, 24, 23, 22, 21, 20)

Aldehyde, ketones $\xrightarrow{rxn}$ HCN or Grignard reagents RMgX

4.) Carbocation stability & GOC effects. (25, 24, 22, 21, 20)
- Hyperconjugation
- Resonance
- Inductive effects

Rank ⇒ $3^\circ > 2^\circ > 1^\circ$ stability

5.) Aldol & cross-aldol Condensation (24, 22, 20)

6.) Ozonolysis of Alkenes (24, 20, 19)

Use O_3 & Zn/H_2O to break a double bond or find original alkenes.

7.) Lassaigne's Test – (25, 23.)

'Blood Red' colour – both N & S present

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8.) Nucleophilic substitution (2024, 2022)

S_N1 loves 3° C — forms Carbocation

S_N2 loves 1° C — results in Inversion.

9.) Elimination (Saytzeff's rule) (24, 21, 20)

10.) Purification Technique (2025, 2024)

└ when to use steam distillation (o & p-nitro phenol) v/s fract^n distillation

11.) Electrophilic Aromatic Substitution (24, 22, 19)

- Halogenation
- Nitration
- Friedel-Craft rxn

] — on benzene ring

12.) Amines – Hinsberg & Carbylamine Tests (21, 20)

Hinsberg ↓ distinguishes 1°, 2°, 3° Amines

Carbylamine ↳ 'foul smell' for 1°

13.) Diazonium salt rxns. (25, 22)

↳ Aniline → Diazonium salt → Iodobenzene or Phenol

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14.) Acidic strength trends : (25, 22)

Phenols & Carboxylic Acids — Comparison

15.) Isomerism ⟶ Metamerism (25, 24, 21)

↳ Counting total no. of isomers etc.

16.) Clemmenson Reduction (2023)

17.) Hoffman-Bromamide degradation (2023)

↳ Amide to→ Amine (1 less Carbon)

18.) Kjeldahl's Method — est. 'N' %. (2022)

↳ Not for Nitro, Azo or ~~Atoi~~ 'N' in ring (pyridine)

19.) Grignard with CO_2 (2022)

20.) Functional Group Reagents (2025, 2024)

[DIBAL-H & PCC]

——— x ——— x ———